# Cars, Planes, Trains

Physics and Mathematics of Sustainable Energy

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A few facts:

- Gasoline: 10 kWh per liter or 38 kWh per gallon
- Average gas mileage for car in US: 25mph, but this ranges considerably.
- Burning one gallon of gasoline releases around 9 kg of $CO_2$.

1. Let's compare driving 1000 miles in conventional and electric vehicles.

   (a) In the conventional car, how much gas does this use?
   (b) How much does this gas cost?
   (c) How much $CO_2$ is emitted by the car?
   (d) How much of the thermal energy released when burning the gasoline goes into the kinetic energy of the car? Assume that the car's engine has an efficiency of 0.25.
   (e) How many kWh of electricity would be needed by an electric car to go 1000 miles. Assume that the efficiency of the electric car is 0.85.
   (f) How much would this electricity cost?
   (g) How much $CO_2$ would be emitted as a result of generating this amount of electricity, assuming the US average carbon intensity.
   (h) How much $CO_2$ would be emitted as a result of generating this amount of electricity if the electricity was generated in a coal-burning power plant with an intensity of 0.82 kg/kWh?

2. Suppose you fly from New York to Oregon twice in a year. What are the emissions associated with these flights. Put this number in perspective.

3. Suppose you want to get from London to Paris.

   (a) What are the emissions if you fly?
   (b) What are the emissions if you drive in a petrol (gas) car?
   (c) What are the emissions is you take the Eurostar train?